

Chapter II

A. Questions

1. Explain the difference between the process and the program.
2. Please describe the main functions and differences of the various schedulers.
3. Please explain what status is usually available for itineraries?
4. Describe the circumstances under which a new itinerary is generated.
5. Please explain the possible reasons for the end of the tour.
6. Describe what information about the trip is typically stored in the Trip Control section.
7. Briefly describe common queues in the operating system and explain their purpose.
8. Please explain what is body switching?
9. When a thread is created, what resources are used? How is it different from creating an itinerary?

B. Calculation problems

1. There is a set of itineraries as follows:

Process CPU burst time (ms)

P1	15
P2	7
P3	4
P4	10

The arrival time is 0 and is in the order of P1, P2, P3 and P4.

Use FCFS and non-preemptive SJF scheduling methods to find the average wait time, respectively.

2. Suppose there are four processes P1, P2, P3, and P4, all of which arrive at time 0, in the order of P1, P2, P3, and P4, and their CPU burst times are as follows:

Process CPU burst time (ms) priority

P1	6	2
P2	5	0
P3	7	3
P4	4	1

- a. When using FCFS, SJF, non-preemptive priority (the lower the number, the higher the priority), and RR (time slice = 3) scheduling methods, please draw a

diagram similar to the text about the order in which the procedures are executed.

- b. What is the average wait time for all trips?
- c. What is the response time for each trip?

3. Suppose there are four processes P1, P2, P3, P4, and P5, all of which arrive at time 0, in the order of P1, P2, P3, P4, and P5, and their CPU burst times are as follows:

Process CPU burst time (ms) priority

P1	10	3
P2	3	1
P3	2	3
P4	1	4
P5	5	2

(a) When using FCFS, SJF, non-preemptive priority (the lower the number, the higher the priority), and RR (time slice = 3, executed in order of arrival) scheduling methods, please draw a diagram similar to the text about the order in which the itinerary is executed.

(b) What is the average wait time for each trip in the various algorithms mentioned above? Which algorithm has the lowest average wait time?

(c) If you want RR to perform better, according to the rule of thumb, what should be the minimum time slicing in the execution case of the above example?